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Models and Methods for a Self-Organising Distributed Computing Platform for Satellite Constellations

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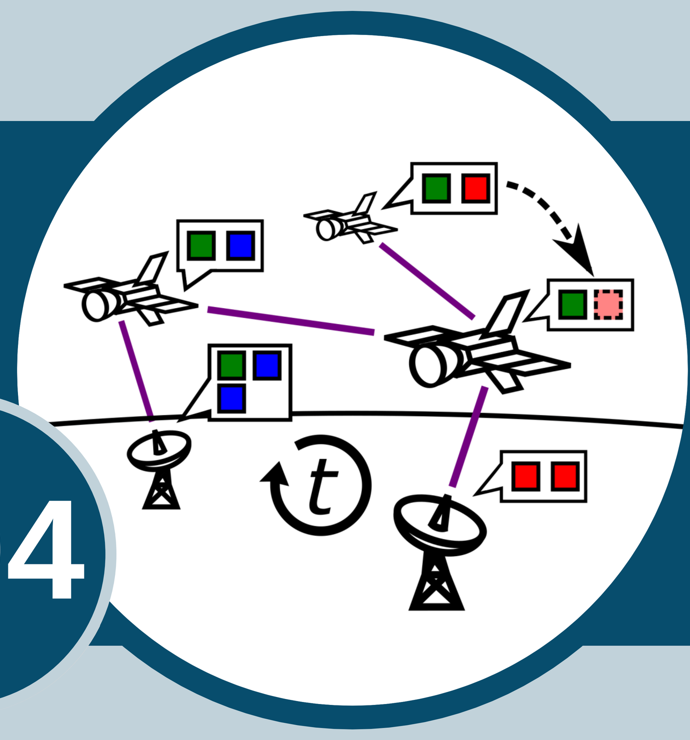


Motivation

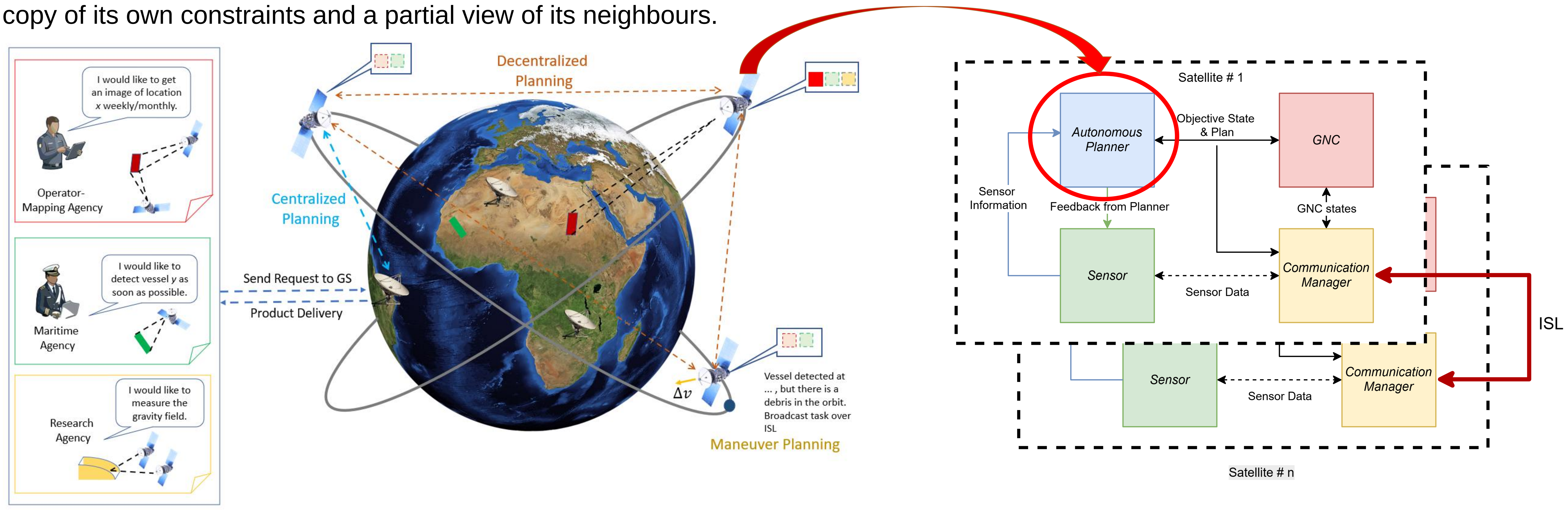
- Multi-satellite missions are typically executed via direct ground commanding of each spacecraft, or through a central hub satellite that relays pre-computed plans to individual detector satellites [1].
- VLEO satellites has limited communication windows and transmission delays that makes centralized control impractical.
- Distributed space systems will take advantage of real-time decision making algorithms to manage mission planning and perform related maneuver planning without human intervention.
- Plug and play technology enables satellites to perform more than one mission.

Objectives

- **Autonomous Decision Making:** Developing algorithms for autonomous decision making in multi-satellite missions, enabling satellites to make decisions based on local information and interactions with neighbouring satellites and ground stations.
- **Decentralized Mission Planning:** Creating decentralized mission planning algorithms that allow satellites to collaboratively plan and execute tasks without relying on a central controller.
- **Maneuver Planning:** Generate optimal maneuver plans online while keeping computational cost low.

Research Highlights

Given a set of incoming user requests *R* and a set of satellites *S* with heterogeneous sensor capabilities and orbital states, find an assignment a: R → S and, for each assigned pair, an executable manoeuvre and observation plan that maximises served requests while respecting Δv, attitude, energy, and link constraints. The problem is solved jointly but in a decentralised way: each satellite owns a local copy of its own constraints and a partial view of its neighbours.



Methods

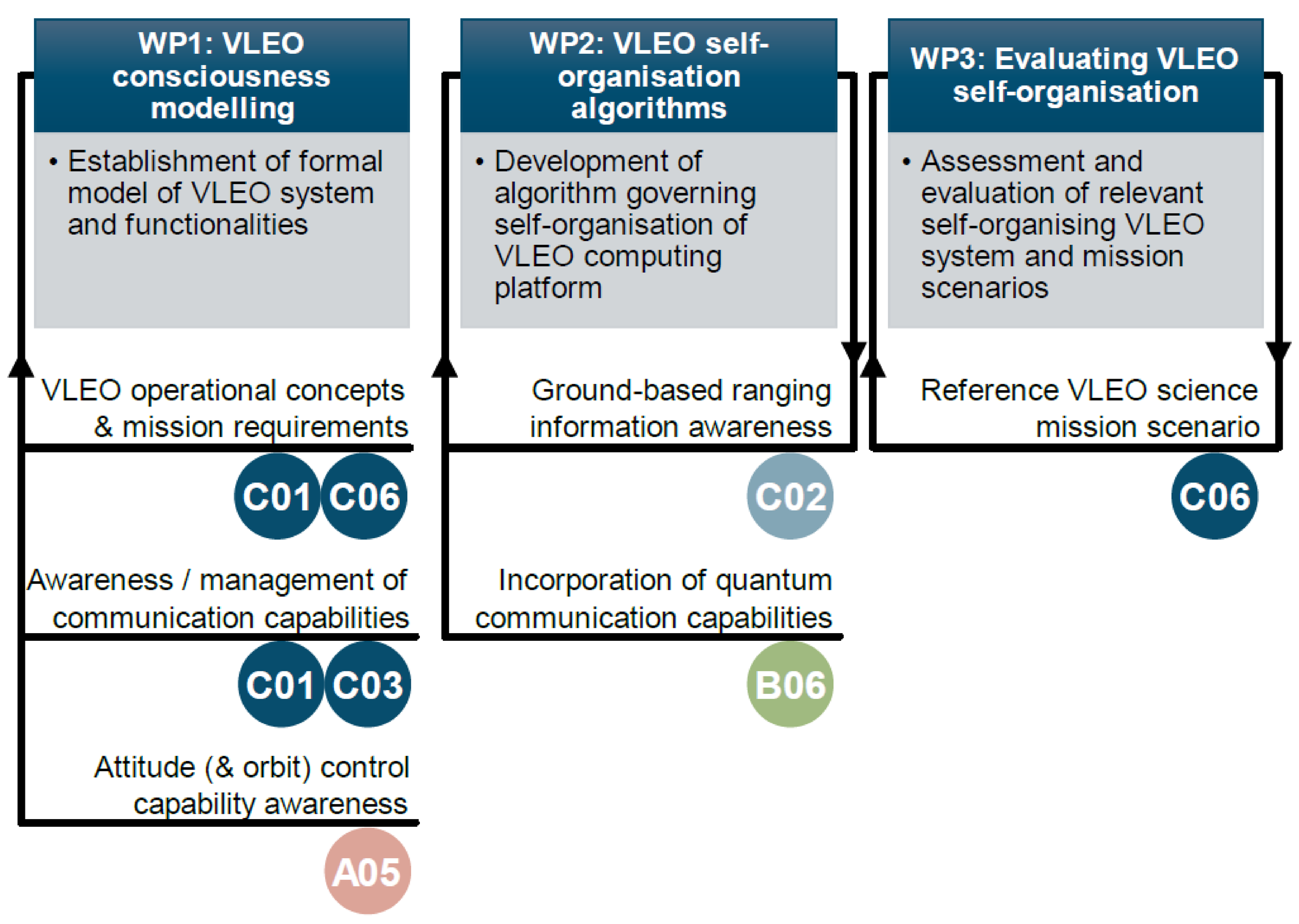
Task allocation (Who takes the task?):

- **LP / MILP**: Optimal assignment under linear constraints [2]. Gives the optimal centralised reference solution used to benchmark decentralised methods.
- **MDP - Dec-POMDP**: A multi-agent decision process under partial observability

Maneuver planning (How satellite executes the task?):

- MDP formulation: state = (orbit, attitude, debris context, energy); actions = sequential slew commands and Δv impulses; reward shapes pointing accuracy, fuel use etc.
- Event Triggered MPC: Satellites can react to updated debris ephemerides, attitude estimates, and revised allocations.

Collaborations



References

[1] Adams, C., Kempa, B., Iatauro, M., Frank, J., \& Vaughan, W. (2023). An overview of distributed spacecraft autonomy at NASA Ames.
[2] Arslan, A., & Annighöfer, B. (2025). Automated Task Allocation Enabling Virtual Satellite Constellation for Vleo: A Mixed Integer Linear Programming Approach. 2025 AIAA DATC/IEEE 44th Digital Avionics Systems Conference (DASC), 1-7.

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